CS2600

THE UNIVERSITY OF WARWICK
April Examinations 2018/19
Algorithms

Time allowed: 2 hours.

Answer **THREE** questions: **ONE** from Section A and **TWO** from Section B.

Make it clear on the cover of the answerbook which three questions you have selected.

Read carefully the instructions on the answer book and make sure that the particulars required are entered on **each** answer book.

Calculators are not required and not permitted.

(continued)

Section A

Answer **ONE** question from this section.

1. (a) **[5 points]** Give the definition of *polynomial-time reduction.*

   (b) **[15 points]** For each of the following statements, indicate whether it is true or false. (Justification is not required. You will be awarded 3 points for each correct answer and -3 points for each incorrect or missing answer. Should the sum of the awarded points be negative, you will receive 0 points for this part of the question.)

   i. Every NP-hard problem is NP-complete.

   ii. Every NP-complete problem is NP-hard.

   iii. As far as we know, it is possible that some, but not all, NP-hard problems have polynomial-time algorithms.

   iv. As far as we know, it is possible that the set cover problem has no polynomial-time algorithm.

   v. As far as we know, it is possible that the maximum flow problem has no polynomial-time algorithm.

2. (a) **[7 points]** Prove that the following *Integer Programming* problem is NP-hard:
   Input: matrix $A \in \mathbb{Z}^{m \times n}$ and vector $\vec{b} \in \mathbb{Z}^m$.
   Output: does there exist a vector $\vec{x} \in \mathbb{Z}^n$ with nonnegative components such that $A \cdot \vec{x} = \vec{b}$?
   You can assume that the Subset Sum problem is NP-hard.

   (b) **[7 points]** Prove that the following *Overlap* problem belongs to the complexity class NP:
   Input: undirected graph G, positive integer k.
   Output: does G contain a clique C of size k and an independent set I of size k that overlap (i.e., $C \cap I \neq \emptyset$)?

   (c) **[6 points]** What is the maximum integer k for which there exists a graph G with 7 vertices such that (G, k) is a positive instance of the *Overlap* problem from part (b)? Justify your answer.

(continued)

Section B

Answer **TWO** questions from this section.

(continued)

3. (a) **[6 points]** Suppose the greedy algorithm for the set cover problem is run on the following instance: the universe is $U = \{a, b, c, d, e, f, g, h\}$ and the sets are $\{a, b, c, d\}$, $\{b, c, d\}$, $\{a, e\}$, $\{b, f\}$, $\{c, g\}$, $\{d, h\}$. What is the size of the cover that the algorithm will output? (If there are multiple outcomes possible, specify all of them.)

   (b) **[5 points]** Let $\mathcal{N} = (G, s, t, c)$ be a network and f a flow in $\mathcal{N}$. Give the definition of the *value* of the flow f.

   (c) **[8 points]** A network $\mathcal{N} = (G, s, t, c)$ is called *acyclic* if the graph G has no cycles. For each integer $n \geq 2019$, give an example of an acyclic network with $O(n)$ nodes and $O(n)$ edges in which there is a cut of capacity 2019 and the number of integral flows of maximum value is $\Omega(2^n)$. (Recall that a flow f is *integral* if $f(e) \in \mathbb{Z}$ for all edges e in G.)

   (d) **[5 points]** Suppose a nondecreasing function $f\colon \mathbb{Z}_{\geq 0} \to \mathbb{Z}_{\geq 0}$ satisfies the inequality $f(n) \leq 5f(\lfloor n/3 \rfloor) + O(n^2)$ for all $n \geq 2$. Prove, without using the master theorem, that $f(n) = O(n^2)$.

Denote by $x \# y$ the maximum of two numbers x and y. For example, $2 \# 3 = 3 \# 2 = 3$ and $3 \# 3 = 3$.

Given an arithmetic expression that uses numbers and operations $+$, $\times$, and $\#$, we can specify in which order the operations should be performed by putting pairs of parentheses in it. For example, $3 \# 2 + 1 \times 4$ may become $((3 \# 2) + 1) \times 4 = 16$. For this expression, there are in total five different ways to put (two pairs of) parentheses in it.

   (e) **[6 points]** What is the *minimum* value of expressions that can be obtained by putting parentheses in $3 \# 2 + 1 \times 4$?

   (f) **[10 points]** Give a polynomial-time algorithm that determines the minimum value of expressions that can be obtained by putting parentheses in a given expression.

   Assume that the expression is given in the following form: the array $A[1..n]$ specifies the numbers (nonnegative integers), left to right, and the array $B[1..n-1]$ specifies the operations. For instance, the expression from part (e) would be given as follows:

$$\begin{array}{lcccc} A[1..4]: & 3 & 2 & 1 & 4 \\ B[1..3]: & \# & + & \times & \end{array}$$

(continued)

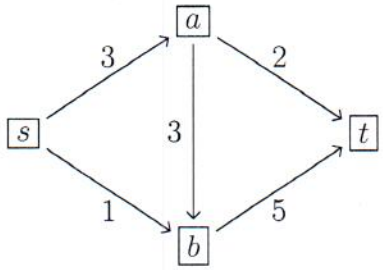
4. (a) **[6 points]** Prove that Quicksort with naive pivot choice has worst-case running time $O(n^2)$.

   (b) **[6 points]** Give the definition of *decision tree for sorting n-element arrays.*

   (c) **[5 points]** Suppose the DFS algorithm is run on the directed graph $G = (V, E)$ with $V = \{a, b, c, d\}$ and $E = \{ab, ac, cb, cd, dc\}$, starting from node a. Assume that, whenever the algorithm has a choice of vertices to explore, it always picks the one that is alphabetically first. Will any back edge be discovered and, if yes, what is the back edge that will be discovered first?

Let $G = (V, E)$ be a directed graph with edge weights $w\colon E \to \mathbb{R}_{>0}$ and π a path in G. Recall that the *length* of the path π is the sum of weights of its edges. Define the *width* of π as the minimum weight of its edges.

   (d) **[6 points]** In the following graph, calculate the width of the shortest path from s to t:



   (e) **[10 points]** Give a polynomial-time algorithm, that, given G, w, and two distinct vertices s and t of G, finds the maximum width of paths from s to t. Justify the correctness of your algorithm and prove that it runs in polynomial time.

   (f) **[7 points]** Give an example of G, w, s, and t ($s \neq t$) for which there is a unique maximum-width path from s to t, and this path is neither a shortest path nor a longest path.

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5. (a) **[8 points]** Let $G = (V, E)$ be an undirected graph with weights $w\colon E \to \mathbb{R}$. Suppose T is a minimum spanning tree of G, and an edge $e \in E$ belongs to the tree T. Prove that there exists a set $S \subseteq V$ such that the edge e is one of the lightest edges that connect S and $V \setminus S$, that is, $u \in S$, $v \in V \setminus S$, and

$$w(e) = \min_{\substack{(u,v)\in E:\\ u\in S,\ v\in V\setminus S}} w(u, v).$$

   (b) **[6 points]** Is the following Boolean formula in CNF satisfiable?

$$(x_1 \vee \overline{x}_2) \,\&\, (\overline{x}_1 \vee x_3) \,\&\, x_2 \,\&\, (\overline{x}_3 \vee \overline{x}_4) \,\&\, (x_1 \vee x_3 \vee x_4)$$

   (c) **[8 points]** For both of the following problems, specify what is given as input and what is the required output: (i) SAT, (ii) Orthogonal Vectors (OV).

   (d) **[6 points]** Formulate a theorem that links algorithms (and their worst-case running times) for the SAT problem and for the OV problem.

   (e) **[6 points]** The theorem that you formulated in part (d) is proved by showing a procedure that transforms instances of SAT into instances of OV. What does this procedure output when run on the CNF from part (b)?

   (f) **[6 points]** Is the output of this procedure on this input CNF a positive instance of OV? Justify your answer.

(end)